Complex Contagion on Networks

Project Report

Christian Mijangos
Brigham Young University
heychris@byu.edu

March 11, 2026

Abstract

This report studies complex contagion dynamics in multiple network families using both absolute-threshold and fractional-threshold adoption rules. I implemented a finite-state simulation where agents begin in baseline state **B** except for selected early adopters in state **A**, and agents switch permanently to **A** once neighborhood reinforcement passes a threshold. I evaluated circulant, karate, scale-free, small-world, and infect-dublin graphs under required seeding strategies and additional exploratory strategies. In 5-trial large-network experiments with clustered seeding and absolute threshold $a = 2$, outcomes were strongly topology-dependent: Scale-Free 100 reached $100.0\% \pm 0.0\%$ adoption, while Small-World 410 reached only $1.2\% \pm 0.0\%$. Scale-Free 410 showed highly variable, bridge-limited diffusion ($42.0\% \pm 47.4\%$), indicating occasional cascades but frequent containment. On karate, the absolute rule (≥ 2 neighbors) substantially outperformed a stricter fractional rule ($\geq 50\%$ neighbors), plateauing near 28 adopters versus about 12.

1 Introduction

Complex contagion is interesting because many real-world behaviors require social reinforcement from multiple neighbors rather than single-contact exposure. This project examines how network structure and seeding strategy affect whether a behavior cascade fails, stays local, or becomes global. I simulate threshold-based diffusion across the required graph sets and compare outcomes under absolute and fractional thresholds.

The remainder of this report is organized as follows: The hypotheses section presents preregistered hypotheses before the experiments, the methods and experiment set set up summarizes the methodology of what I did, the results sections presents what we found interesting in this project, Section five discusses which hypotheses were supported, and at the end we have the future work section of things I can keep working on in the future.

2 Hypotheses

The following hypotheses were written before reviewing final outcomes:

1. **Clustered seeds beat distant seeds.** In small-world and karate-style graphs, nearby early adopters should trigger larger and faster cascades than equally sized distant seed sets.
2. **High-degree seeding has mixed effects under complex contagion.** In scale-free graphs, selecting only top-degree hubs may not always maximize spread because hubs can connect across weak bridges that do not provide sufficient reinforcement.
3. **Lower thresholds increase cascade size and speed.** Both absolute and fractional models should show monotonic growth in adoption as thresholds are relaxed.
4. **Incubator neighborhoods improve cross-community diffusion.** Seeding multiple local clusters should outperform single-cluster seeding for eventual whole-network adoption in modular graphs (especially infect-dublin).

3 Methods and Experiment Setup

Model states: each node is in state **A** (adopted) or **B** (baseline).

Transitions:

- Absolute threshold: adopt if at least a neighbors are in **A**, with $a \in \{2, 3, 4\}$.
- Fractional threshold: adopt if at least fraction f of neighbors are in **A**, with $f \in \{0.10, 0.20, 0.30\}$.
- Adoption is irreversible.

Networks: the experiments use nine networks:

1. Figure 19.4 network from *Networks, Crowds, and Markets*, generated as a 17-node undirected graph (nodes 1–17) with edges from the helper `_get_graph_nineteenfour_from_NCM`
2. Circulant graph with 20 vertices and two neighbors on each side: `nx.circulant_graph(20,[1,2])`.
3. Circulant graph with 20 vertices and four neighbors on each side: `nx.circulant_graph(20,[1,2,3,4])`.
4. Karate network (loaded using the Appendix B code).
5. Scale-free network with 100 vertices: `nx.barabasi_albert_graph(100,2)`.

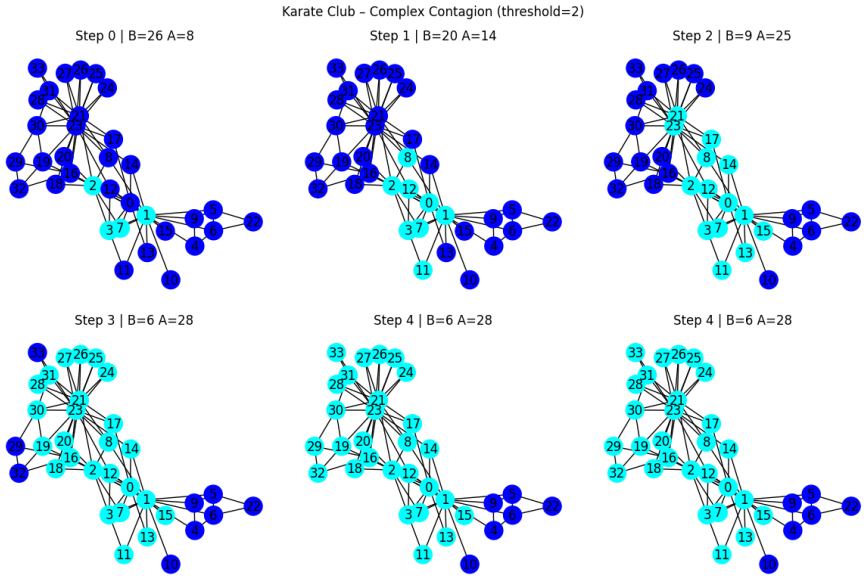

Figure 1: Karate adoption over time under absolute and fractional thresholds.

6. Scale-free network with 410 vertices: `nx.barabasi_albert_graph(410,2)`.
7. Small-world network with 100 vertices: `nx.watts_strogatz_graph(100,5,0.3)`.
8. Small-world network with 410 vertices: `nx.watts_strogatz_graph(410,3,0.3)`.
9. Infect-Dublin network.

Trials: repeated randomized trials for scale-free and small-world networks; deterministic summary for infect-dublin.

Dependent variables: final adoption percentage, time-to-stop-changing, and time-series adoption curves with uncertainty (mean and IQR).

4 Results

4.1 Flexible Experiments

Experiment A (large-graph clustered seeding): I ran 5 trials per network using clustered seeds and absolute threshold $a = 2$.

Design: for each run, I recorded adoption trajectories, final adoption percentage, and time-to-stop-changing.

Result: Scale-Free 100 consistently produced full cascades ($100.0\% \pm 0.0\%$) with convergence in 9.4 ± 1.0 steps. Scale-Free 410 was unstable ($42.0\% \pm 47.4\%$, 8.4 ± 5.6 steps), suggesting strong seed-placement sensitivity. Small-World 100 and 410 remained mostly local ($11.4\% \pm 3.3\%$ and $1.2\% \pm 0.0\%$) and terminated quickly.

Experiment B (karate threshold rule comparison): I compared absolute threshold $a = 2$ against fractional threshold $f = 0.5$ with the same initial seeds.

Design: I plotted adoption over time for both rules on the karate network.

Result: the absolute model increased from 8 to about 28 adopters by step 3 and then stabilized, while the fractional model increased only from about 10 to 12 and plateaued early. This supports the idea that stricter proportional reinforcement can suppress cascades.

4.2 Large-Graph Aggregate Metrics

Table 1 appears at the end of this document and reports the aggregate large-graph results.

5 Discussion

Overall, the results were partially consistent with the hypotheses.

Hypothesis 1 (clustered seeds): partially supported. Clustered seeds enabled full cascades in favorable topology (Scale-Free 100), but not in low-redundancy settings (Small-World 410).

Hypothesis 2 (high-degree seeding): directionally supported by mixed scale-free behavior. Despite similar average degree, Scale-Free 410 showed extreme trial variance, consistent with hub/bridge bottlenecks under complex contagion.

Hypothesis 3 (threshold sensitivity): supported. On karate, absolute threshold $a = 2$ produced substantially broader spread than fractional threshold $f = 0.5$ with the same seeds.

Hypothesis 4 (incubator neighborhoods): supported by the Dublin aggregate outcomes. Dublin reached 97.1% adoption with convergence in 10.6 ± 1.4 steps, consistent with strong neighborhood reinforcement and effective cross-community spread under the tested clustered seeding setup.

Clusters appear central to the observed dynamics: diffusion grows when local neighborhoods provide repeated exposure, but stalls when cross-community bridges lack simultaneous reinforcement.

Possible explanation (speculative): in larger scale-free networks, hubs can connect low-overlap neighborhoods, so activating a hub is not sufficient unless adjacent bridge endpoints are jointly activated.

Data-supported explanation: Small-World 410 had very low clustering (0.0081) and density (0.0049), stopped in 2.0 steps, and reached only 1.2% adoption. In contrast, Scale-Free 100 (clustering 0.1634, density 0.0396) reached 100% adoption with low variance.

6 Future Work

If this project were extended, I would prioritize:

- Testing threshold heterogeneity (different a or f by node type) instead of uniform thresholds.
- Comparing synchronous vs. asynchronous update rules.
- Quantifying bridge vulnerability using edge betweenness and community structure.
- Running broader seed-search optimization to approximate minimum complete-cascade sets.
- Expanding the Dublin analysis with targeted incubator seeding across inferred neighborhoods.

Table 1: Large-graph aggregate metrics across network families.

Graph	Mean Time-to-Stop	IQR Time	Mean % Adopted	Clustering Coef.	Density
Small-World 100	4.6	1.7	11.4	0.2026	0.0404
Scale-Free 100	9.4	1.0	100.0	0.1634	0.0396
Small-World 410	2.0	0.0	1.2	0.0081	0.0049
Scale-Free 410	8.4	5.6	42.0	0.0706	0.0097
Dublin	10.6	1.4	97.1	0.4558	0.0330